THE COMPOSITION OF TWO DERIVED TENSOR FUNCTORS IS A DERIVED TENSOR FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-compose-tensor-functors-general-algebra (source lines 5958-5975); the proof environment begins on the second line after it and is its own proof, closing at source line 6064.

Editorial note. The statement and the proof invoke six author results of the same chapter by label; all six are retained as context spans and the generator expands the context transitively (budget 800 lines for this card), while reference targets in other chapters that carry canonical project tags are auto-declared as non-load-bearing source pointers.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: the composition of two derived tensor functors is isomorphic to a derived tensor functor given by the tensor product of the two bimodules.

Allowed standard prerequisites (editorial).

- Differential graded algebras and bimodules. Dg R-algebras (A,d), (B,d), (C,d), and dg (A,B)- and (B,C)-bimodules, with their homotopy and derived categories D(-,d).. It is the setting of the statement (source lines 5960-5964).
- Derived tensor product of bimodules. The left derived functor - tensor^L_A N on D(A,d) defined by the construction of the chapter, together with its functoriality in the bimodule up to quasi-isomorphism.. The two functors being composed and the announced N" are derived tensor functors (source lines 5965-5974, 5978-5982).
- Property (P) for bimodules and its filtration sequences. Every dg bimodule has a quasi-isomorphic representative with property (P), carrying an admissible filtration whose graded pieces are direct sums of shifts of the free bimodule, and a split admissible short exact sequence displaying the filtration (the author's lemmas lemma-bimodule-resolve and lemma-bimodule-property-P-sequence, retained as context).. Both reduction chains of the proof run along these filtrations (source lines 5980-5982, 5992-6024, 6035-6051).

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Exactness of the derived tensor functor. The derived tensor functor is exact as a functor of triangulated categories, commutes with direct sums, and transforms admissible exact sequences of complexes into admissible exact sequences (the author's lemma-compose-tensor-functors-general, retained as context).. It is what licenses the two reductions (source lines 5988-6018).

Source reference. Original lemma 00DB: sources/stacks-chow/tags.txt, source lines 491–491. Cross-chapter pointer whose target carries the canonical tag 00DB in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 06XN: sources/stacks-chow/tags.txt, source lines 8857–8857. Cross-chapter pointer whose target carries the canonical tag 06XN in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09IZ: sources/stacks-chow/tags.txt, source lines 12057–12057. Cross-chapter pointer whose target carries the canonical tag 09IZ in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1464–1464. Cross-chapter pointer whose target carries the canonical tag 0149 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09J0: sources/stacks-chow/tags.txt, source lines 12058–12058. Cross-chapter pointer whose target carries the canonical tag 09J0 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 014C: sources/stacks-chow/tags.txt, source lines 1467–1467. Cross-chapter pointer whose target carries the canonical tag 014C in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 013N: sources/stacks-chow/tags.txt, source lines 1442–1442. Cross-chapter pointer whose target carries the canonical tag 013N in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

$$(A \otimes_R B)_B \otimes^{\mathbf{L}}_B {}_B(B \otimes_R C)_C \longrightarrow (A \otimes_R B \otimes_R C)_C \tag{0.0.1}$$

Lemma 0.1. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let $f : N \to N'$ be a homomorphism of differential graded (A, B)-bimodules. Then f induces a morphism of functors*

$$1 \otimes f : - \otimes^{\mathbf{L}}_A N \longrightarrow - \otimes^{\mathbf{L}}_A N'$$

If f is a quasi-isomorphism, then $1 \otimes f$ is an isomorphism of functors.

Proof. Let M be a differential graded A-module with property (P). Then $1 \otimes f : M \otimes_A N \to M \otimes_A N'$ is a map of differential graded B-modules. Moreover, this is functorial with respect to M. Since the functors $- \otimes^{\mathbf{L}}_A N$ and $- \otimes^{\mathbf{L}}_A N'$ are computed

by tensoring on objects with property (P) (Lemma 0.5) we obtain a transformation of functors as indicated.

Assume that f is a quasi-isomorphism. Let $F_\bullet$ be the given filtration on M. Observe that $M \otimes_A N = \operatorname{colim} F_i(M) \otimes_A N$ and $M \otimes_A N' = \operatorname{colim} F_i(M) \otimes_A N'$. Hence it suffices to show that $F_n(M) \otimes_A N \to F_n(M) \otimes_A N'$ is a quasi-isomorphism (filtered colimits are exact, see Algebra, Lemma 00DB). Since the inclusions $F_n(M) \to F_{n+1}(M)$ are split as maps of graded A-modules we see that

$$0 \to F_n(M) \otimes_A N \to F_{n+1}(M) \otimes_A N \to F_{n+1}(M)/F_n(M) \otimes_A N \to 0$$

is a short exact sequence of differential graded B-modules. There is a similar sequence for N' and f induces a map of short exact sequences. Hence by induction on n (starting with $n = -1$ when $F_{-1}(M) = 0$) we conclude that it suffices to show that the map $F_{n+1}(M)/F_n(M) \otimes_A N \to F_{n+1}(M)/F_n(M) \otimes_A N'$ is a quasi-isomorphism. This is true because $F_{n+1}(M)/F_n(M)$ is a direct sum of shifts of A and the result is true for $A[k]$ as $f : N \to N'$ is a quasi-isomorphism. □

Lemma 0.2. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let M be a differential graded (A, B)-bimodule. There exists a homomorphism $P \to M$ of differential graded (A, B)-bimodules which is a quasi-isomorphism such that P has property (P) as defined above.*

Proof. Immediate from Lemmas 0.6 and 0.7. □

Lemma 0.3. *Let R be a ring. Let (A, d), (B, d), and (C, d) be differential graded R-algebras. Let N be a differential graded (A, B)-bimodule. Let N' be a differential graded (B, C)-module. Assume (0.7.1) is an isomorphism. Then the composition*

$$D(A, d) \xrightarrow{-\otimes^{\mathbf{L}}_A N} D(B, d) \xrightarrow{-\otimes^{\mathbf{L}}_B N'} D(C, d)$$

is isomorphic to $- \otimes^{\mathbf{L}}_A N''$ with $N'' = N \otimes_B N'$ viewed as (A, C)-bimodule.

Proof. Let us define a transformation of functors

$$(- \otimes^{\mathbf{L}}_A N) \otimes^{\mathbf{L}}_B N' \longrightarrow - \otimes^{\mathbf{L}}_A N''$$

To do this, let M be a differential graded A-module with property (P). According to the construction of the functor $- \otimes^{\mathbf{L}}_A N''$ of the proof of Lemma 0.5 the plain tensor product $M \otimes_A N''$ represents $M \otimes^{\mathbf{L}}_A N''$ in $D(C, \mathrm{d})$. Then we write

$$M \otimes_A N'' = M \otimes_A (N \otimes_B N') = (M \otimes_A N) \otimes_B N'$$

The module $M \otimes_A N$ represents $M \otimes^{\mathbf{L}}_A N$ in $D(B, \mathrm{d})$. Choose a quasi-isomorphism $Q \to M \otimes_A N$ where Q is a differential graded B-module with property (P). Then $Q \otimes_B N'$ represents $(M \otimes^{\mathbf{L}}_A N) \otimes^{\mathbf{L}}_B N'$ in $D(C, \mathrm{d})$. Thus we can define our map via

$$(M \otimes^{\mathbf{L}}_A N) \otimes^{\mathbf{L}}_B N' = Q \otimes_B N' \to M \otimes_A N \otimes_B N' = M \otimes^{\mathbf{L}}_A N''$$

The construction of this map is functorial in M and compatible with distinguished triangles and direct sums; we omit the details. Consider the property T of objects M of $D(A, \mathrm{d})$ expressing that this map is an isomorphism. Then

(1) if T holds for M_i then T holds for $\bigoplus M_i$,
(2) if T holds for 2-out-of-3 in a distinguished triangle, then it holds for the third, and

(3) T holds for $A[k]$ because here we obtain a shift of the map (0.7.1) which we have assumed is an isomorphism.

Thus by Remark 0.8 property T always holds and the proof is complete. □

Lemma 0.4. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let P be a differential graded (A, B)-bimodule having property (P) with corresponding filtration $F_\bullet$, then we obtain a short exact sequence*

$$0 \to \bigoplus F_i P \to \bigoplus F_i P \to P \to 0$$

of differential graded (A, B)-bimodules which is split as a sequence of graded (A, B)-bimodules.

Proof. Immediate from Lemmas 0.6 and 0.9. □

Lemma 0.5. *In the situation above, the left derived functor of F exists. We denote it $- \otimes^{\mathbf{L}}_A N : D(A, d) \to D(B, d)$.*

Proof. We will use Derived Categories, Lemma 06XN to prove this. As our collection $\mathcal{P}$ of objects we will use the objects with property (P). Property (1) was shown in Lemma 0.7. Property (2) holds because if $s : P \to P'$ is a quasi-isomorphism of modules with property (P), then s is a homotopy equivalence by Lemma 0.10. □

Lemma 0.6. *Let R be a ring. Let (A, d) and (B, d) be differential graded algebras over R. The construction above defines an equivalence of categories*

$$\begin{array}{c} \textit{differential graded} \\ (A, B)\textit{-bimodules} \end{array} \longleftrightarrow \begin{array}{c} \textit{right differential graded} \\ A^{opp} \otimes_R B\textit{-modules} \end{array}$$

Proof. Immediate from discussion the above. □

Lemma 0.7. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $P \to M$ of differential graded A-modules such that*

(1) *$P \to M$ is a quasi-isomorphism, and*
(2) *P has property (P).*

Proof. Set $M = M_0$. We inductively choose short exact sequences

$$0 \to M_{i+1} \to P_i \to M_i \to 0$$

where the maps $P_i \to M_i$ are chosen as in Lemma 0.11. This gives a "resolution"

$$\ldots \to P_2 \xrightarrow{f_2} P_1 \xrightarrow{f_1} P_0 \to M \to 0$$

Then we set

$$P = \bigoplus\nolimits_{i \geq 0} P_i$$

as an A-module with grading given by $P^n = \bigoplus_{a+b=n} P^b_{-a}$ and differential (as in the construction of the total complex associated to a double complex) by

$$\mathrm{d}_P(x) = f_{-a}(x) + (-1)^a \mathrm{d}_{P_{-a}}(x)$$

for $x \in P^b_{-a}$. With these conventions P is indeed a differential graded A-module. Recalling that each P_i has a two step filtration $0 \to P'_i \to P_i \to P''_i \to 0$ we set

$$F_{2i} P = \bigoplus\nolimits_{i \geq j \geq 0} P_j \subset \bigoplus\nolimits_{i \geq 0} P_i = P$$

and we add P'_{i+1} to $F_{2i}P$ to get F_{2i+1}. These are differential graded submodules and the successive quotients are direct sums of shifts of A. By Lemma 0.12 we see that the inclusions $F_iP \to F_{i+1}P$ are admissible monomorphisms. Finally, we have to show that the map $P \to M$ (given by the augmentation $P_0 \to M$) is a quasi-isomorphism. This follows from Homology, Lemma 09IZ. □

$$N_B \otimes^{\mathbf{L}}_B N' \longrightarrow (N \otimes_B N')_C \tag{0.7.1}$$

Remark 0.8. Let R be a ring. Let (A, d) be a differential graded R-algebra. Using P-resolutions we can sometimes reduce statements about general objects of $D(A, \mathrm{d})$ to statements about $A[k]$. Namely, let T be a property of objects of $D(A, \mathrm{d})$ and assume that

(1) if K_i, $i \in I$ is a family of objects of $D(A, \mathrm{d})$ and $T(K_i)$ holds for all $i \in I$, then $T(\bigoplus K_i)$,
(2) if $K \to L \to M \to K[1]$ is a distinguished triangle of $D(A, \mathrm{d})$ and T holds for two, then T holds for the third object, and
(3) $T(A[k])$ holds for all $k \in \mathbf{Z}$.

Then T holds for all objects of $D(A, \mathrm{d})$. This is clear from Lemmas 0.9 and 0.7.

Lemma 0.9. *Let (A, d) be a differential graded algebra. Let P be a differential graded A-module. If $F_\bullet$ is a filtration as in property (P), then we obtain an admissible short exact sequence*

$$0 \to \bigoplus F_iP \to \bigoplus F_iP \to P \to 0$$

of differential graded A-modules.

Proof. The second map is the direct sum of the inclusion maps. The first map on the summand F_iP of the source is the sum of the identity $F_iP \to F_iP$ and the negative of the inclusion map $F_iP \to F_{i+1}P$. Choose homomorphisms $s_i : F_{i+1}P \to F_iP$ of graded A-modules which are left inverse to the inclusion maps. Composing gives maps $s_{j,i} : F_jP \to F_iP$ for all $j > i$. Then a left inverse of the first arrow maps $x \in F_jP$ to $(s_{j,0}(x), s_{j,1}(x), \ldots, s_{j,j-1}(x), 0, \ldots)$ in $\bigoplus F_iP$. □

Lemma 0.10. *Let (A, d) be a differential graded algebra. Let M and N be differential graded A-modules.*

(1) *Let $P \to M$ be a P-resolution as in Lemma 0.7. Then*
$$\mathrm{Hom}_{D(A,d)}(M, N) = \mathrm{Hom}_{K(Mod_{(A,d)})}(P, N)$$
(2) *Let $N \to I$ be an I-resolution as in Lemma 0.15. Then*
$$\mathrm{Hom}_{D(A,d)}(M, N) = \mathrm{Hom}_{K(Mod_{(A,d)})}(M, I)$$

Proof. Let $P \to M$ be as in (1). Since $P \to M$ is a quasi-isomorphism we see that

$$\mathrm{Hom}_{D(A,\mathrm{d})}(P, N) = \mathrm{Hom}_{D(A,\mathrm{d})}(M, N)$$

by definition of the derived category. A morphism $f : P \to N$ in $D(A, \mathrm{d})$ is equal to $s^{-1}f'$ where $f' : P \to N'$ is a morphism and $s : N \to N'$ is a quasi-isomorphism. Choose a distinguished triangle

$$N \to N' \to Q \to N[1]$$

As s is a quasi-isomorphism, we see that Q is acyclic. Thus $\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(P, Q[k]) = 0$ for all k by Lemma 0.14. Since $\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(P, -)$ is cohomological, we conclude that we can lift $f' : P \to N'$ uniquely to a morphism $f : P \to N$. This finishes the proof.

The proof of (2) is dual to that of (1) using Lemma 0.13 in stead of Lemma 0.14. □

Lemma 0.11. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $P \to M$ of differential graded A-modules with the following properties*

1. *$P \to M$ is surjective,*
2. *$\mathrm{Ker}(d_P) \to \mathrm{Ker}(d_M)$ is surjective, and*
3. *P sits in an admissible short exact sequence $0 \to P' \to P \to P'' \to 0$ where P', P'' are direct sums of shifts of A.*

Proof. Let P_k be the free A-module with generators x, y in degrees k and $k+1$. Define the structure of a differential graded A-module on P_k by setting $\mathrm{d}(x) = y$ and $\mathrm{d}(y) = 0$. For every element $m \in M^k$ there is a homomorphism $P_k \to M$ sending x to m and y to $\mathrm{d}(m)$. Thus we see that there is a surjection from a direct sum of copies of P_k to M. This clearly produces $P \to M$ having properties (1) and (3). To obtain property (2) note that if $m \in \mathrm{Ker}(\mathrm{d}_M)$ has degree k, then there is a map $A[k] \to M$ mapping 1 to m. Hence we can achieve (2) by adding a direct sum of copies of shifts of A. □

Lemma 0.12. *Let (A, d) be a differential graded algebra. Let $M \to P$ be a surjective homomorphism of differential graded A-modules. If P is projective as a graded A-module, then $M \to P$ is an admissible epimorphism.*

Proof. This is immediate from the definitions. □

Lemma 0.13. *Let (A, d) be a differential graded algebra. Let I be a differential graded A-module with property (I). Then*

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(N, I) = 0$$

for all acyclic differential graded A-modules N.

Proof. We will use that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a triangulated category (Proposition 0.18). Let $F_\bullet$ be a filtration on I as in property (I). The short exact sequence of Lemma 0.17 produces a distinguished triangle. Hence by Derived Categories, Lemma 0149 it suffices to show that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(N, I/F_iI) = 0$$

for all acyclic differential graded A-modules N and all i. Each of the differential graded modules I/F_iI has a finite filtration by admissible monomorphisms, whose graded pieces are products of $A^\vee[k]$. Thus it suffices to prove that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(N, A^\vee[k]) = 0$$

for all acyclic differential graded A-modules N and all k. This follows from Lemma 0.16 and the fact that $(-)^\vee$ is an exact functor. □

Lemma 0.14. *Let (A, d) be a differential graded algebra. Let P be a differential graded A-module with property (P). Then*

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(P, N) = 0$$

for all acyclic differential graded A-modules N.

Proof. We will use that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a triangulated category (Proposition 0.18). Let $F_\bullet$ be a filtration on P as in property (P). The short exact sequence of Lemma 0.9 produces a distinguished triangle. Hence by Derived Categories, Lemma 0149 it suffices to show that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(F_iP, N) = 0$$

for all acyclic differential graded A-modules N and all i. Each of the differential graded modules F_iP has a finite filtration by admissible monomorphisms, whose graded pieces are direct sums of shifts $A[k]$. Thus it suffices to prove that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(A[k], N) = 0$$

for all acyclic differential graded A-modules N and all k. This follows from Lemma 0.19. □

Lemma 0.15. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $M \to I$ of differential graded A-modules such that*

(1) *$M \to I$ is a quasi-isomorphism, and*
(2) *I has property (I).*

Proof. Set $M = M_0$. We inductively choose short exact sequences

$$0 \to M_i \to I_i \to M_{i+1} \to 0$$

where the maps $M_i \to I_i$ are chosen as in Lemma 0.20. This gives a "resolution"

$$0 \to M \to I_0 \xrightarrow{f_0} I_1 \xrightarrow{f_1} I_1 \to \ldots$$

Denote I the differential graded A-module with graded parts

$$I^n = \prod\nolimits_{i \geq 0} I_i^{n-i}$$

and differential defined by

$$\mathrm{d}_I(x) = f_i(x) + (-1)^i \mathrm{d}_{I_i}(x)$$

for $x \in I_i^{n-i}$. With these conventions I is indeed a differential graded A-module. Recalling that each I_i has a two step filtration $0 \to I_i' \to I_i \to I_i'' \to 0$ we set

$$F_{2i}I^n = \prod\nolimits_{j \geq i} I_j^{n-j} \subset \prod\nolimits_{i \geq 0} I_i^{n-i} = I^n$$

and we add a factor I_{i+1}' to $F_{2i}I$ to get $F_{2i+1}I$. These are differential graded submodules and the successive quotients are products of shifts of $A^\vee$. By Lemma 0.21 we see that the inclusions $F_{i+1}I \to F_iI$ are admissible monomorphisms. Finally, we have to show that the map $M \to I$ (given by the augmentation $M \to I_0$) is a quasi-isomorphism. This follows from Homology, Lemma 09J0. □

Lemma 0.16. *Let (A, d) be a differential graded algebra. Then we have*

$$\mathrm{Hom}_{Mod_{(A,d)}}(M, A^\vee[k]) = \mathrm{Ker}(d : (M^\vee)^k \to (M^\vee)^{k+1})$$

and

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(M, A^\vee[k]) = H^k(M^\vee)$$

as functors in the differential graded A-module M.

Proof. This is clear from the discussion above. □

Lemma 0.17. *Let (A, d) be a differential graded algebra. Let I be a differential graded A-module. If $F_\bullet$ is a filtration as in property (I), then we obtain an admissible short exact sequence*

$$0 \to I \to \prod I/F_i I \to \prod I/F_i I \to 0$$

of differential graded A-modules.

Proof. Omitted. Hint: This is dual to Lemma 0.9. □

Proposition 0.18. *Let (A, d) be a differential graded algebra. The homotopy category $K(Mod_{(A,d)})$ of differential graded A-modules with its natural translation functors and distinguished triangles is a triangulated category.*

Proof. We know that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a pre-triangulated category. Hence it suffices to prove TR4 and to prove it we can use Derived Categories, Lemma 014C. Let $K \to L$ and $L \to M$ be composable morphisms of $K(\mathrm{Mod}_{(A,\mathrm{d})})$. By Lemma 0.22 we may assume that $K \to L$ and $L \to M$ are admissible monomorphisms. In this case the result follows from Lemma 0.23. □

Lemma 0.19. *Let (A, d) be a differential graded algebra. Then we have*

$$\mathrm{Hom}_{Mod_{(A,d)}}(A[k], M) = \mathrm{Ker}(d : M^{-k} \to M^{-k+1})$$

and

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(A[k], M) = H^{-k}(M)$$

for any differential graded A-module M.

Proof. Immediate from the definitions. □

Lemma 0.20. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $M \to I$ of differential graded A-modules with the following properties*

1. *$M \to I$ is injective,*
2. *$\mathrm{Coker}(d_M) \to \mathrm{Coker}(d_I)$ is injective, and*
3. *I sits in an admissible short exact sequence $0 \to I' \to I \to I'' \to 0$ where I', I'' are products of shifts of $A^\vee$.*

Proof. We will use the functors $N \mapsto N^\vee$ (from left to right differential graded modules and from right to left differential graded modules) constructed in Section 1 and all of their properties. For every $k \in \mathbf{Z}$ let Q_k be the free left A-module with generators x, y in degrees k and $k+1$. Define the structure of a left differential graded A-module on Q_k by setting $\mathrm{d}(x) = y$ and $\mathrm{d}(y) = 0$. Arguing exactly as in the proof of Lemma 0.11 we find a surjection

$$\bigoplus\nolimits_{i \in I} Q_{k_i} \longrightarrow M^\vee$$

of left differential graded A-modules. Then we can consider the injection

$$M \to (M^\vee)^\vee \to (\bigoplus\nolimits_{i \in I} Q_{k_i})^\vee = \prod\nolimits_{i \in I} I_{k_i}$$

where $I_k = Q_{-k}^\vee$ is the "dual" right differential graded A-module. Further, the short exact sequence $0 \to A[-k-1] \to Q_k \to A[-k] \to 0$ produces a short exact sequence $0 \to A^\vee[k] \to I_k \to A^\vee[k+1] \to 0$.

The result of the previous paragraph produces $M \to I$ having properties (1) and (3). To obtain property (2), suppose $\overline{m} \in \mathrm{Coker}(\mathrm{d}_M)$ is a nonzero element of degree

k. Pick a map $\lambda : M^k \to \mathbf{Q}/\mathbf{Z}$ which vanishes on $\mathrm{Im}(M^{k-1} \to M^k)$ but not on m. By Lemma 0.16 this corresponds to a homomorphism $M \to A^\vee[k]$ of differential graded A-modules which does not vanish on m. Hence we can achieve (2) by adding a product of copies of shifts of $A^\vee$. □

Lemma 0.21. *Let (A, d) be a differential graded algebra. Let $I \to M$ be an injective homomorphism of differential graded A-modules. If I is graded injective, then $I \to M$ is an admissible monomorphism.*

Proof. This is immediate from the definitions. □

Lemma 0.22. *Let (A, d) be a differential graded algebra. Let $L_1 \to L_2 \to \ldots \to L_n$ be a sequence of composable homomorphisms of differential graded A-modules. There exists a commutative diagram*

$$\begin{array}{ccccccc} L_1 & \longrightarrow & L_2 & \longrightarrow & \ldots & \longrightarrow & L_n \\ \uparrow & & \uparrow & & & & \uparrow \\ M_1 & \longrightarrow & M_2 & \longrightarrow & \ldots & \longrightarrow & M_n \end{array}$$

in $Mod_{(A,d)}$ such that each $M_i \to M_{i+1}$ is an admissible monomorphism and each $M_i \to L_i$ is a homotopy equivalence.

Proof. The case $n = 1$ is without content. Lemma 1.1 is the case $n = 2$. Suppose we have constructed the diagram except for M_n. Apply Lemma 1.1 to the composition $M_{n-1} \to L_{n-1} \to L_n$. The result is a factorization $M_{n-1} \to M_n \to L_n$ as desired. □

Lemma 0.23. *Let (A, d) be a differential graded algebra. Suppose that $\alpha : K \to L$ and $\beta : L \to M$ are admissible monomorphisms of differential graded A-modules. Then there exist distinguished triangles $(K, L, Q_1, \alpha, p_1, d_1)$, $(K, M, Q_2, \beta \circ \alpha, p_2, d_2)$ and $(L, M, Q_3, \beta, p_3, d_3)$ for which TR4 holds.*

Proof. Say $\pi_1 : L \to K$ and $\pi_3 : M \to L$ are homomorphisms of graded A-modules which are left inverse to α and β. Then also $K \to M$ is an admissible monomorphism with left inverse $\pi_2 = \pi_1 \circ \pi_3$. Let us write Q_1, Q_2 and Q_3 for the cokernels of $K \to L$, $K \to M$, and $L \to M$. Then we obtain identifications (as graded A-modules) $Q_1 = \mathrm{Ker}(\pi_1)$, $Q_3 = \mathrm{Ker}(\pi_3)$ and $Q_2 = \mathrm{Ker}(\pi_2)$. Then $L = K \oplus Q_1$ and $M = L \oplus Q_3$ as graded A-modules. This implies $M = K \oplus Q_1 \oplus Q_3$. Note that $\pi_2 = \pi_1 \circ \pi_3$ is zero on both Q_1 and Q_3. Hence $Q_2 = Q_1 \oplus Q_3$. Consider the commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & K & \to & L & \to & Q_1 & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & K & \to & M & \to & Q_2 & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & L & \to & M & \to & Q_3 & \to & 0 \end{array}$$

The rows of this diagram are admissible short exact sequences, and hence determine distinguished triangles by definition. Moreover downward arrows in the diagram above are compatible with the chosen splittings and hence define morphisms of triangles

$$(K \to L \to Q_1 \to K[1]) \longrightarrow (K \to M \to Q_2 \to K[1])$$

and

$$(K \to M \to Q_2 \to K[1]) \longrightarrow (L \to M \to Q_3 \to L[1]).$$

Note that the splittings $Q_3 \to M$ of the bottom sequence in the diagram provides a splitting for the split sequence $0 \to Q_1 \to Q_2 \to Q_3 \to 0$ upon composing with $M \to Q_2$. It follows easily from this that the morphism $\delta : Q_3 \to Q_1[1]$ in the corresponding distinguished triangle

$$(Q_1 \to Q_2 \to Q_3 \to Q_1[1])$$

is equal to the composition $Q_3 \to L[1] \to Q_1[1]$. Hence we get a structure as in the conclusion of axiom TR4. □

1. Injective modules and differential graded algebras

Lemma 1.1. *Let (A, d) be a differential graded algebra. Let $\alpha : K \to L$ be a homomorphism of differential graded A-modules. There exists a factorization*

$$K \xrightarrow{\tilde{\alpha}} \tilde{L} \xrightarrow{\pi} L, \quad K \xrightarrow{\alpha} L$$

in $Mod_{(A,d)}$ such that

1. *$\tilde{\alpha}$ is an admissible monomorphism (see Definition 1.2),*
2. *there is a morphism $s : L \to \tilde{L}$ such that $\pi \circ s = id_L$ and such that $s \circ \pi$ is homotopic to $id_{\tilde{L}}$.*

Proof. The proof is identical to the proof of Derived Categories, Lemma 013N. Namely, we set $\tilde{L} = L \oplus C(1_K)$ and we use elementary properties of the cone construction. □

Definition 1.2. Let (A, d) be a differential graded algebra.

1. A homomorphism $K \to L$ of differential graded A-modules is an *admissible monomorphism* if there exists a graded A-module map $L \to K$ which is left inverse to $K \to L$.
2. A homomorphism $L \to M$ of differential graded A-modules is an *admissible epimorphism* if there exists a graded A-module map $M \to L$ which is right inverse to $L \to M$.
3. A short exact sequence $0 \to K \to L \to M \to 0$ of differential graded A-modules is an *admissible short exact sequence* if it is split as a sequence of graded A-modules.

Lemma 1.3. *Let R be a ring. Let (A, d), (B, d), and (C, d) be differential graded R-algebras. Assume that (0.0.1) is an isomorphism. Let N be a differential graded (A, B)-bimodule. Let N' be a differential graded (B, C)-bimodule. Then the composition*

$$D(A, d) \xrightarrow{-\otimes^{\mathbf{L}}_A N} D(B, d) \xrightarrow{-\otimes^{\mathbf{L}}_B N'} D(C, d)$$

is isomorphic to $- \otimes^{\mathbf{L}}_A N''$ for a differential graded (A, C)-bimodule N'' described in the proof.

Proof. By Lemma 0.1 we may replace N and N' by quasi-isomorphic bimodules. Thus we may assume N, resp. N' has property (P) as differential graded (A, B)-bimodule, resp. (B, C)-bimodule, see Lemma 0.2. We claim the lemma holds with the (A, C)-bimodule $N'' = N \otimes_B N'$. To prove this, it suffices to show that

$$N_B \otimes_B^{\mathbf{L}} N' \longrightarrow (N \otimes_B N')_C$$

is an isomorphism in $D(C, \mathrm{d})$, see Lemma 0.3.

Let $F_\bullet$ be the filtration on N as in property (P) for bimodules. By Lemma 0.4 there is a short exact sequence

$$0 \to \bigoplus F_i N \to \bigoplus F_i N \to N \to 0$$

of differential graded (A, B)-bimodules which is split as a sequence of graded (A, B)-bimodules. A fortiori this is an admissible short exact sequence of differential graded B-modules and this produces a distinguished triangle

$$\bigoplus F_i N_B \to \bigoplus F_i N_B \to N_B \to \bigoplus F_i N_B[1]$$

in $D(B, \mathrm{d})$. Using that $- \otimes_B^{\mathbf{L}} N'$ is an exact functor of triangulated categories and commutes with direct sums and using that $- \otimes_B N'$ transforms admissible exact sequences into admissible exact sequences and commutes with direct sums we reduce to proving that

$$(F_p N)_B \otimes_B^{\mathbf{L}} N' \longrightarrow (F_p N)_B \otimes_B N'$$

is a quasi-isomorphism for all p. Repeating the argument with the short exact sequences of (A, B)-bimodules

$$0 \to F_p N \to F_{p+1} N \to F_{p+1} N / F_p N \to 0$$

which are split as graded (A, B)-bimodules we reduce to showing the same statement for $F_{p+1} N / F_p N$. Since these modules are direct sums of shifts of $(A \otimes_R B)_B$ we reduce to showing that

$$(A \otimes_R B)_B \otimes_B^{\mathbf{L}} N' \longrightarrow (A \otimes_R B)_B \otimes_B N'$$

is a quasi-isomorphism.

Choose a filtration $F_\bullet$ on N' as in property (P) for bimodules. Choose a quasi-isomorphism $P \to (A \otimes_R B)_B$ of differential graded B-modules where P has property (P). We have to show that $P \otimes_B N' \to (A \otimes_R B)_B \otimes_B N'$ is a quasi-isomorphism because $P \otimes_B N'$ represents $(A \otimes_R B)_B \otimes_B^{\mathbf{L}} N'$ in $D(C, \mathrm{d})$ by the construction in Lemma 0.5. As $N' = \operatorname{colim} F_p N'$ we find that it suffices to show that $P \otimes_B F_p N' \to (A \otimes_R B)_B \otimes_B F_p N'$ is a quasi-isomorphism. Using the short exact sequences $0 \to F_p N' \to F_{p+1} N' \to F_{p+1} N' / F_p N' \to 0$ which are split as graded (B, C)-bimodules we reduce to showing $P \otimes_B F_{p+1} N' / F_p N' \to (A \otimes_R B)_B \otimes_B F_{p+1} N' / F_p N'$ is a quasi-isomorphism for all p. Then finally using that $F_{p+1} N' / F_p N'$ is a direct sum of shifts of ${}_B(B \otimes_R C)_C$ we conclude that it suffices to show that

$$P \otimes_B {}_B(B \otimes_R C)_C \to (A \otimes_R B)_B \otimes_B {}_B(B \otimes_R C)_C$$

is a quasi-isomorphism. Since $P \to (A \otimes_R B)_B$ is a resolution by a module satisfying property (P) this map of differential graded C-modules represents the morphism (0.0.1) in $D(C, \mathrm{d})$ and the proof is complete. □